

CoRoT-2b

R-2032

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_121003 · created 2026-08-08T01:23:46+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456203.6325 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.112 +/- 0.045
Transit depth (Rp/Rs)^2	1.26 +/- 1.0 [%]
Orbital Inclination (inc)	87.91 +/- 1.97
Airmass coefficient 1 (a1)	0.99 +/- 0.018
Airmass coefficient 2 (a2)	0.005 +/- 0.025
Scatter in the residuals of the lightcurve fit is	1.8 %
Best Comparison Star	#1 - [312, 297]
Optimal Aperture	2.86
Optimal Annulus	12.66
Transit Duration (day)	0.068 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.112 ± 0.045 vs 0.1667 (deviation 1.22σ)
Duration fit vs published	0.068 d vs 0.095 d (deviation 7.71σ)
Mid-time O-C	-6.6 min
Depth SNR	2.5
Residual scatter	1.8 %

Light curve

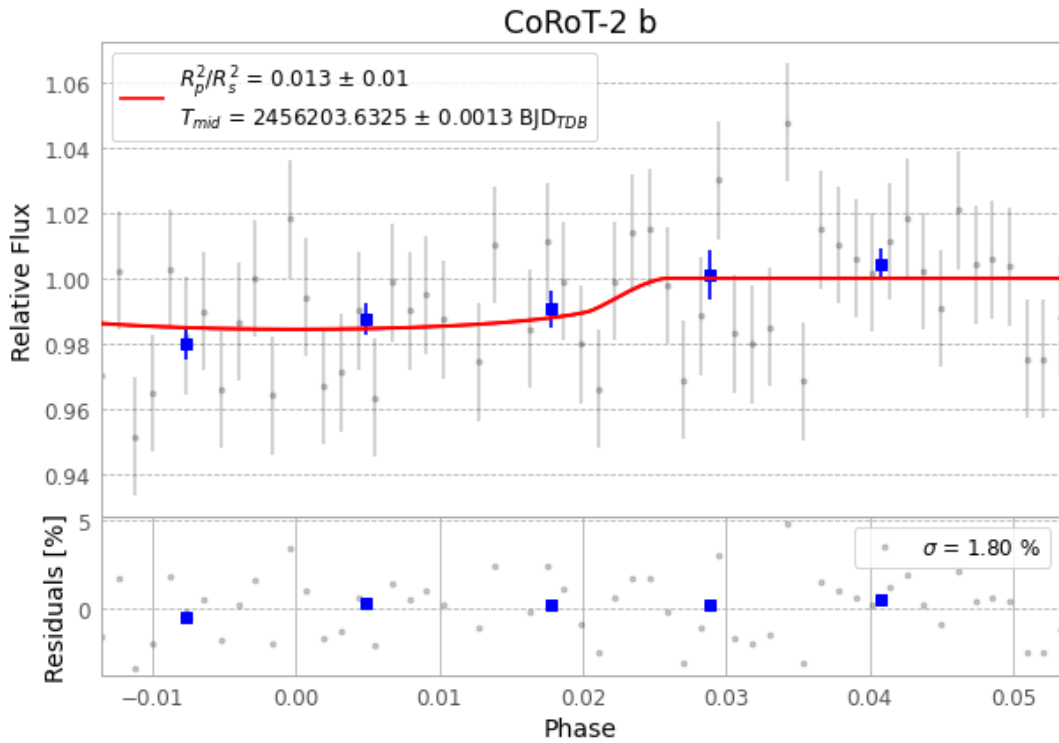


Figure 1 — FinalLightCurve_CoRoT-2 b_2012-10-02.png (SHA-256 329cca836d0aea59...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [366, 271], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 20ad84f02b4a2beb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

57 FITS frames (37 MB), CoRoT-2121003023313.FITS → CoRoT-2121003052112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-121003.log (de7f44ad7a1c...), FinalLightCurve_CoRoT-2 b_2012-10-02.png (329cca836d0a...), FinalLightCurve_CoRoT-2 b_2012-10-02.csv (d7d1ec208076...)

Compute resources

Wall time 117.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 01:23Z.